CSC311: Introduction to Machine Learning

Project Report

Submitted by

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1 Executive Summary

We propose to classify a sport as soccer, baseball, or basketball using survey responses about its accessibility, playing characteristics, and cultural associations. The dataset contains 870 responses from 290 respondents, with 290 examples of each sport. We will compare regularized softmax regression, random forests, and Naive Bayes to investigate how linear relationships, nonlinear interactions, and textual evidence contribute to classification. We hypothesize that softmax regression will provide a strong balance between predictive performance and interpretability when trained on encoded survey responses and TF-IDF text features. We will select the final model using validation results grouped by respondent and assess it using accuracy and macro-F1. A projected test accuracy will be reported after completing these experiments.

2 Data Exploration

Dataset Overview and Splitting

- **Dataset Composition**: The dataset contains 870 survey responses collected from 290 distinct students, with each respondent evaluating all three sports (Soccer, Baseball, and Basketball).
- **Class Balance**: The target labels are perfectly balanced (290 observations per sport overall; exactly 203 per sport in the 70% training split). No class rebalancing or oversampling is needed.
- **Respondent Isolation (`id`)**: Each respondent's unique identifier is excluded from the feature matrix X and used exclusively for grouped partitioning. Keeping all responses from the same participant within the same split prevents the model from memorizing respondent-specific vocabulary, bias, or stylistic quirks.
- **Strict Zero-Leakage Split**: The 290 respondent IDs are partitioned into:
  - **Training split (70%)**: 203 respondents (609 observations) used for exploratory analysis, distribution inspection, and parameter estimation.
  - **Validation split (15%)**: 43 respondents (129 observations) reserved strictly for hyperparameter tuning.
  - **Internal test split (15%)**: 44 respondents (132 observations) locked away untouched until final model evaluation.

  All statistical parameters (imputation medians, standard scaling means/stds, and text vocabularies) are learned strictly on the 70% training split to eliminate data snooping.

Feature Preprocessing Pipeline

Table 1 summarizes the 12 survey features, identifier handling, target labels, and machine learning rationales.

Table 1: Comprehensive breakdown of all 14 survey features, preprocessing pipeline, and machine learning rationales.

Survey Feature	Raw Type	Preprocessing Pipeline	Final Representation	Machine Learning Rationale & Advantage
Respondent ID (`id`)	Categorical ID	Excluded from features; used for grouped partition	Grouping Index (not in X)	Prevents style memorization; ensures zero leakage across folds.
Ease of Playing (`difficulty_rating`)	Numeric (0–10)	Mean imputation ($\mu = 5.48$); Z-score standardization	Standardized continuous	Eliminates scale disparity; stabilizes gradient descent convergence.
Strategy Importance (`strategy_importance`)	5-point Likert	Monotonic mapping (1–5); neutral imputation (3)	Ordinal scalar $\in \{1..5\}$	Preserves ordered agreement degrees without inflating dimensionality.
Injury Risk Comfort (`injury_risk_comfort`)	5-point Likert	Monotonic mapping (1–5); neutral imputation (3)	Ordinal scalar $\in \{1..5\}$	Monotonically captures contact vs non-contact sport perception.
Individual Brilliance (`individual_brilliance`)	5-point Likert	Monotonic mapping (1–5); neutral imputation (3)	Ordinal scalar $\in \{1..5\}$	Differentiates star-driven play (basketball) from collective team play.
Playing Locations (`play_location`)	Multi-select CSV	Lowercase; split by delimiter; binary multi-hot expansion	Binary vector (7 dims)	Accommodates concurrent, non-exclusive venues (e.g., school and park).
Toronto Seasons (`play_season`)	Multi-select CSV	Lowercase; split seasons; binary multi-hot expansion	Binary vector (4 dims)	Models multi-season playability (e.g., summer/fall outdoor vs winter).
Music Genres (`associated_music`)	Multi-select CSV	Lowercase; split genres; binary multi-hot expansion	Binary vector (11 dims)	Captures non-exclusive cultural affinities across musical genres.
Foods and Drinks (`associated_foods_drinks`)	Free-form text	Lowercase; strip punctuation; TF-IDF downweighting; missing flag	Sparse TF-IDF + missing flag	**TF-IDF**: Penalizes generic filler ("the", "with"); boosts terms like "popcorn".
Associated Colour (`associated_color`)	Free-form text	Lowercase; clean punctuation; TF-IDF representation; missing flag	TF-IDF vector + missing flag	Unifies casing ("Orange" vs "orange"); isolates discriminative colors.
Associated Song (`associated_song`)	Free-form text	Lowercase; clean punctuation; TF-IDF representation; missing flag	Sparse TF-IDF + missing flag	Captures anthem tokens ("waka", "lights"); absence signals obscure sports.
Emotional Response (`emotional_response`)	Free-form text	Lowercase; strip punctuation; TF-IDF vectorization; missing flag	Sparse TF-IDF + missing flag	Differentiates excitement vs boredom while filtering non-informative words.
Willingness to Pay (`acceptable_price`)	Unstructured text	Regex strip "$", commas; convert "thousand"; median fill; $\log(1+x)$	Log continuous scalar	Compresses skewness ($+10.5 \rightarrow -1.15$), removing leverage of extreme outliers.
Target Sport (`label`)	Class string	Map Soccer $\rightarrow 0$, Baseball $\rightarrow 1$, Basketball $\rightarrow 2$	Integer label $y \in \{0, 1, 2\}$	Standard multi-class target; perfectly balanced (203 per class in train split).

Key Preprocessing Strategies and Rationale

- **TF-IDF vs. Raw Word Counts**: In free-text survey responses, uninformative stopwords such as "the", "a", and "with" appear frequently across all three sports. A simple term-frequency counter would erroneously assign these generic words the highest weights. TF-IDF applies an inverse-document-frequency penalty, heavily discounting common words while amplifying discriminative, sport-specific terms (e.g., "pitcher", "popcorn", "court").
- **Handling Extreme Outliers with** $\log(1 + x)$: Ticket price willingness exhibits extreme positive skewness (+10.5), spanning from \$0 to \$10,000. Applying $\log(1 + \text{price})$ compresses this long tail, transforming the distribution into a near-symmetric curve (skewness -1.15) and preventing outlier respondents from distorting linear model weights.
- **Informative Missingness Indicators**: Blank responses range from 0.8% (7 entries) to 8.7% across features, with the song question exhibiting the highest vacancy (76 of 870 entries, or 90 entries / 10.3% when including text placeholders such as "none" or "n/a"). Rather than discarding sparse text rows, a binary indicator (`is_missing`) is appended to each text feature, allowing models to leverage deliberate omission as a predictive signal (e.g., respondents unable to name a song for less mainstream sports).

3 Methodology

Data splitting and validation We reserve 15% of respondent IDs as an internal test set, 15% as a validation set, and 70% for model training. On the remaining respondents, we will use five-fold grouped cross-validation for hyperparameter tuning. All three responses from each respondent will remain together throughout this process. Imputation values, scaling parameters, categorical vocabularies, and text vocabularies will be learned separately from each fold's training data. The internal test set will remain unused during feature selection and hyperparameter tuning. Model 1: Regularized softmax regression Softmax regression will predict the probabilities of the three sports from a combination of structured features and TF-IDF text features. It is appropriate as a relatively simple model for a dataset with many potentially sparse text features. L2 regularization will help control overfitting. We will tune the regularization parameter, compare unigrams with unigrams plus bigrams, and vary the minimum document frequency used to retain text features. We will inspect model coefficients to identify which responses contribute most strongly to each prediction. Model 2: Random forest A random forest will model nonlinear relationships and interactions be-

tween responses. For example, the relationship between a season response and a sport may depend on the locations selected by the respondent. We will first evaluate the forest using structured features, then test whether adding a restricted text vocabulary improves validation performance. We will tune maximum tree depth, minimum samples per leaf, and the number of features considered at each split. Candidate settings could include 100 and 300 trees, maximum depths of 5, 10, and unrestricted depth, and minimum leaf sizes of 1, 3, and 5. Model 3: Naive Bayes Multinomial Naive Bayes will provide a probabilistic text baseline using nonnegative word-count or TF-IDF features from the free-text questions. It is computationally inexpensive and can combine many weak word-level signals. We will tune the smoothing parameter over values such as 0.1, 0.5, 1, and 2, and compare word counts with TF-IDF. Its conditional-independence assumption may limit performance when words or survey responses are strongly related. Fair comparison and feature experiments All model families will use the same respondent-based folds and receive hyperparameter tuning before comparison. We will compare structured-only, text-only, and combined representations where appropriate to determine how much each feature group contributes. The supplied kNN implementation can serve as an additional reference baseline. We will also report a constant-class baseline, which achieves 33.3% accuracy on this balanced dataset. Evaluation We will report accuracy, macro-F1, per-class precision and recall, and confusion matrices. Accuracy measures overall correctness, while macro-F1 gives each sport equal importance and helps reveal uneven performance across classes. We will summarize the mean and standard deviation of cross-validation scores. Our primary selection metric will be mean validation accuracy, with macro-F1 and consistency across folds informing close comparisons.

4 Results

For the proposal, use planned analysis rather than invented scores:

We will summarize the tuned models' cross-validation performance in the following table:

TABLE

We will analyze confusion matrices to determine which sports are most frequently confused. We will also examine whether errors are associated with missing responses, uncommon song titles, ambiguous text, or similar structured answers.

We hypothesize that softmax regression will perform well because regularization can help it handle sparse text features. Random forests may perform better if interactions among structured responses are especially informative. Naive Bayes will establish how effectively word-level evidence alone distinguishes the classes.

After selecting the model and preprocessing configuration, we will evaluate the selected pipeline once on the internal test set. We will use this result, supported by cross-validation stability, to justify a single projected accuracy for the unseen course test set. No defensible numerical estimate is available yet.

5 Contributions and Learning

This section should contain a list of 3–4 bullets, one for each student. Each student should write at most three sentences describing their main contributions and the most important lesson learned.

6 References

Use references whenever appropriate (e.g., when you consult textbooks, research papers, online tutorials, or external code). Cite in the text using `\cite{}`. For example, let's cite the two references provided in the `.bib` file [1, 2].

References

[1] Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.

[2] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, 2009.

appendices.

A Guiding Questions for Data Exploration

These questions support Section 2. Work through them in order, and record your answers for our dataset in that section rather than here. This appendix is part of the template instructions, so comment it out before you submit, as you do with the rest of the instructions. what is the number of classes?

1. Determine Input to Model
   (a) For each example, what features do we have? What are the types of the features (numerical, categorical, or text)?

   (b) In the released data set, each row has a unique ID identifying which student provided the data. Why should we not use this unique ID as a feature? highly skewed? Is there enough variation? Are there outliers?

2. Data Splitting
   (a) Why is it a good idea to split the provided data into a training set and a test set?

   (b) Why is it a good idea to do the training and test set splitting based on the unique ID identifying each respondent?

   (c) What is the role of the validation set and why do we need this third split?

   (d) Why should we only perform data exploration on the training set and not on the validation or the test sets?

3. Feature Distributions One reason for exploring the distributions of features is to get to know the data before building any model. This exploration can help us catch common issues such as outliers, skewed distributions, class imbalance, missing values, and large difference in feature scales.
   (a) What should we do when exploring the distributions of the features?

   (b) Suppose a feature takes the same value for almost every respondent. Why is this feature worth finding, and what would you do about it?

   (c) Suppose two features are highly correlated, such as two Likert questions that respondents tend to answer the same way. How would you test whether two features are highly correlated, and why does a highly correlated pair cause trouble?

4. Class Imbalance
   (a) What is class imbalance? Why is class imbalance problematic?

   (b) Does the dataset have a class imbalance issue? If so, how severe?

   (c) What are some common strategies to address class imbalance?

5. Missing Values
   (a) First, determine if the dataset contains missing values. Report the percentage of values missing for each feature in a table. A missing value can take many forms such as a null value, an empty string, an empty list, “n/a”, or “-”. Normalize all of these to a single missing marker before you count, or you will underestimate how much is missing.

   (b) Next, for each feature with a substantial missing rate, try to determine why the values are likely missing. Different explanations may require different fixes. What are some possible explanations for missing values?

   (c) Since you never see the missing value, it may not be possible to verify which explanation holds. However, it would be useful to determine whether the missingness of a feature is related to the label or not. If the missingness is related to the label, a missing value itself provides a predictive signal. What is a common strategy for determining this?

   (d) What are some common strategies to handle missing values? One is called imputation.

   (e) When is it appropriate to use each strategy above to handle missing values?

   (f) Suppose that you decide to impute the median value for a numeric feature. Why is it problematic to do the imputation on the validation and test sets?

   (g) Finally, decide how missing values should be handled for each feature in our dataset, and record your decision.

6. Outliers
   (a) What observation might cause you to suspect that certain values are outliers?

   (b) How do I determine whether a feature value should be considered an outlier?

   (c) What are some strategies to handle outliers?

7. Feature Engineering
   (a) How can we convert the responses to a multi-select question into a feature?

   (b) The survey has a few Likert-scale questions that ask respondents to indicate their level of agreement to a statement using an ordered set of options (e.g., from Strongly Disagree to Strongly Agree). We can convert the Likert-scale responses to features by modeling them as ordinal values or one-hot encoded categories. What are some advantages and disadvantages of each approach?

   (c) At least one free-text question asks for a number. How would you turn such a text response into a number, and when would you declare a response unparseable?

   (d) What are some strategies to convert the free-text responses to numeric features?

(e) One common technique for processing the free-text responses is TF-IDF vectorization. Explain this technique in detail. Why is this technique suitable for processing the free-text data?

(f) How should we handle a word that appears at prediction time but never in the training split?

(g) After encoding all the features, you recorded that there are 1100 features and 300 examples. How would the ratio between these two inform your model choices? Why would you choose a simpler model with strong regularization over a larger model?

8. Feature Scaling

(a) Two common choices for feature scaling are standardization and min-max scaling. Explain each in one sentences. Why do we need to perform feature scaling for our dataset, which has Likert-scale, multi-hot encoded, and word count features?

(b) Is each model below affected by feature scaling? Briefly explain.

- K-nearest neighbours
- Decision trees and random forest
- Softmax regression trained using gradient descent
- L2 Regularization
- Naive Bayes

(c) You decide to standardize a feature. Which split do you use to compute the mean and standard deviation, and why?

9. Which Features Are Predictive Next, we would like to determine whether a feature is useful by looking at the feature together with the label. This analysis could also tell you which model families are worth trying and help you justify your model choices in the Methodology section.

(a) How do you examine the relationship between a Likert-scale or numeric feature and the label?

(b) How do you examine the relationship between a multi-select or free-text feature and the label?

(c) Why should you only do this analysis on the training split and not on the validation or test splits?

(d) How can you delete whether a feature is worth keeping by comparing the difference in mean feature values between classes with the spread of feature values within a class? Consider a Likert-scale feature on a scale of 1 to 5. Is the feature worth keeping if two classes have means 3.5 and 3.8 while the standard deviation of answers within a class is 1.2? What about if two classes have means 2.1 and 4.3 while the standard deviation of answers within a class is 0.6?

(e) What does the shape of the relationship tell you about which model families to try? Consider the three scenarios below. Discuss the model families you would try for each one.
   i. The classes separate well along one feature at a threshold.
   ii. The middle values of a Likert feature belong to one class while both extremes belong to another.
   iii. After vectorizing the free-text responses, you have several hundred word-count features. No single word separates the classes, but each word in a response carries a small amount of evidence.

(f) Finally, write down the features you expect to be the most predictive, and name the plot or the number that led you to each one.

B Guiding Questions for Methodology

These questions support Section 3. Work through them in order, and record your answers for our dataset in that section rather than here. This appendix is part of the template instructions, so comment it out before you submit, as you do with the rest of the instructions.

1. Choosing Model Families
   (a) Based on the size of the data set, should we consider fine-tuning a large language model or starting with simpler models? What are some advantages and disadvantages of each approach?

   (b) Which findings from your data exploration would support choosing softmax regression? Which findings would support choosing a decision tree instead?

2. Training Details
   (a) Suppose you train a multilayer perceptron. Which choices would another student need to know to reproduce your results exactly?

   (b) Your softmax regression model reaches 95% accuracy on the training set but only 70% on the validation set. What does this gap tell you, and which regularization or early stopping strategies could you try?

3. Validation
   (a) What is cross validation? If we want to use cross validation, why should we use grouped cross validation, grouped by the unique ID?

   (b) Pre-processing steps such as normalization and text vectorization learn values from the data, like a mean or a vocabulary. Which part of the data should these values be computed from during cross validation, and why?

4. Hyperparameter Tuning
   (a) Which hyperparameters would you tune for a decision tree, and how would you choose the range of values to try?

(b) You try maximum depths from 2 to 10 for a decision tree, and the validation accuracy is highest at a depth of 10. Is your search finished? What would you do next?

(c) After choosing your final model and its hyperparameters, should you retrain it on the training and validation sets combined? What do you gain, and what do you lose?

5. Comparing Model Families Fairly
   (a) With default hyperparameters, softmax regression reaches 78% validation accuracy and a decision tree reaches 72%. You tune softmax regression up to 82% and choose it as your final model. What is wrong with this process, and how would you fix it?

   (b) Across five folds, model A has a mean validation accuracy 1% higher than model B, but the accuracy of each model varies by about 3% from fold to fold. Can you conclude that model A is better? What else could help you decide?

6. Evaluation Metrics
   (a) Suppose one class makes up 60% of the training set. What accuracy does a model that always predicts this class achieve? Which metric would reveal that this model is useless?

   (b) How do you compute precision, recall, and F1 for a 3-class problem? What is the difference between macro-averaging and weighted averaging, and when would you prefer each one?

7. Additional Techniques
   (a) Suppose you apply PCA to several hundred word-count features. How would you decide how many components to keep, and which part of the data should you fit PCA on?

   (b) How could you use a clustering method such as k-means to explore the free- text responses? How would you tell whether the clusters improve your predictions?